

13 listopada - I kolokwium

Ostatnia/przedostatnia środa semestru - II kolokwium
(21/28 stycznia)

EZ

$$a_n = \sqrt[n]{13^n} \quad b_n = \sqrt[n]{7^n + 9^n + 13^n} \quad c_n = \sqrt[n]{13^n + 13^n + 13^n}$$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \sqrt[n]{7^n + 9^n + 13^n}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt[n]{13^n} = 13$$

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} \sqrt[n]{13^n + 13^n + 13^n} \rightarrow \sqrt[n]{3 \cdot 13^n} \rightarrow \sqrt[n]{3} \cdot \sqrt[n]{13^n} \rightarrow 1 \cdot 13 \rightarrow 13$$

Na podstawie tw. o trzech ciągach, skoro

$$a_n \leq b_n \leq c_n \text{ i } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = 13, \text{ to}$$

$$\lim_{n \rightarrow \infty} b_n = 13.$$

$$\lim_{n \rightarrow \infty} \frac{\cos(n! + 2n)}{n^3} = \lim_{n \rightarrow \infty} \left(\frac{1}{n^3} \cdot \cos(n! + 2n) \right) = 0$$

ciąg og.
 $\forall n \in \mathbb{N} \quad \cos(n! + 2n) \leq 1$

$$\lim_{n \rightarrow \infty} \frac{n + (-1)^n}{n - (-1)^n} \rightarrow \frac{n(1 + \frac{(-1)^n}{n})}{n(1 - \frac{(-1)^n}{n})} \rightarrow 1$$

EZ / KOL

ZT: $x_n \rightarrow 0$, $y_n \rightarrow \pm \infty$, $x_n \cdot y_n \rightarrow g \in (-\infty; \infty)$

$$\lim_{n \rightarrow \infty} \left(\frac{n^2 + 5n}{n^2 + 4n + 2} \right)^{3n^2 + 7} = [1^\infty] = \lim_{n \rightarrow \infty} \left(\frac{n^2 + 4n + 2 + n - 2}{n^2 + 4n + 2} \right)^{3n^2 + 7} = \lim_{n \rightarrow \infty} \left(1 + \frac{n - 2}{n^2 + 4n + 2} \right)^{3n^2 + 7} =$$

$$= \left\{ \begin{array}{l} x_n = \frac{n - 2}{n^2 + 4n + 2} \rightarrow 0 \\ y_n = 3n^2 + 7 \rightarrow \infty \end{array} \right. \quad x_n \cdot y_n = \frac{(n - 2)(3n^2 + 7)}{n^2 + 4n + 2} = \frac{3n^3 + 7n - 6n^2 - 14}{n^2 + 4n + 2} \rightarrow \infty \quad \left. \vphantom{\frac{(n - 2)(3n^2 + 7)}{n^2 + 4n + 2}} \right\} = e^\infty$$

Zad. 1

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{2n} = \lim_{n \rightarrow \infty} \left(\frac{n+1-1}{n+1} \right)^{2n} = \lim_{n \rightarrow \infty} \left(1 + \frac{-1}{n+1} \right)^{2n} =$$

$$= \left\{ \begin{array}{l} x_n = \frac{-1}{n+1} \rightarrow 0 \\ y_n = 2n \rightarrow \infty \end{array} \right. \quad x_n \cdot y_n = \frac{-1}{n+1} \cdot 2n = \frac{-2n}{n+1} \rightarrow -2 \quad \left. \vphantom{\lim_{n \rightarrow \infty}} \right\} = e^{-2} = \frac{1}{e^2}$$

Zad. 2

$$\lim_{n \rightarrow \infty} \left(\frac{4n+7}{4n+12} \right)^{-n^2-3} \rightarrow \left(\frac{4n+12-5}{4n+12} \right)^{-n^2-3} \rightarrow \left(1 + \frac{-5}{4n+12} \right)^{-n^2-3} \rightarrow \left\{ \begin{array}{l} x_n = \frac{-5}{4n+12} \rightarrow 0 \\ y_n = -n^2-3 \rightarrow -\infty \end{array} \right.$$

$$x_n \cdot y_n = \frac{-5(-n^2-3)}{4n+12} = \frac{5n^2+15}{4n+12} \rightarrow \infty \quad \left. \vphantom{\lim_{n \rightarrow \infty}} \right\} \rightarrow e^{\infty}$$

Zad. 3

$$a_n = (-1)^n \cdot \frac{2n+1}{5n-1}$$

T_w

Jeśli $\lim_{n \rightarrow \infty} a_n = g$ to każdy podciąg ciągu a_n też ma granicę g

$$\left. \begin{array}{l} a_1, a_2, a_3, \dots, a_n, \dots \rightarrow \lim a_n \\ a_1, a_3, a_5, \dots \\ a_2, a_4, a_8, a_{16}, \dots \end{array} \right\} \text{podciąg } a_n$$

$$a_1, a_3, \dots, a_{2n+1}, \dots \quad \lim_{n \rightarrow \infty} a_{2n+1} = -\frac{2}{5}$$

$$a_2, a_4, \dots, a_{2n}, \dots \quad \lim_{n \rightarrow \infty} a_{2n} = \frac{2}{5}$$

Skoro podciągi a_{2n+1} i a_{2n} mają różne granice, to ciąg a_n nie ma granicy c.n.w.

$$\lim_{n \rightarrow \infty} \frac{\log_4(n+2)}{\log_8(n+2)} = \left[\frac{\infty}{\infty} \right] \rightarrow \frac{\log(n+2) 8}{\log(n+2) 4} \rightarrow \log_4 8 \rightarrow \frac{3}{2}$$

$$(\ln x)' = \frac{1}{x}$$

$$(e^x)' = e^x$$

$$(2^x)' = 2^x \ln 2$$

Zad. 4

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2 n!}{(n+2)!} \rightarrow \frac{(n+1)^2 \cdot n!}{(n+1)(n+2)n!} \rightarrow \frac{(n+1)}{(n+2)} \rightarrow 1$$

Zad. 5

$$\lim_{n \rightarrow \infty} \frac{(n+1)! + n!}{(n+1)! - n!} = \lim_{n \rightarrow \infty} \frac{n!(n+1) + n!}{n!(n+1) - n!} = \lim_{n \rightarrow \infty} \frac{n!(n+1+1)}{n!(n+1-1)} = 1$$